

NOVA PROGRAM

Nova Environmental and Mechanical Requirements

Domain: Environmental qualification

Revision A | Synthetic extraction benchmark

2. Functional requirements

REQ-NVA-ENV-001 | Ingress protection

Connector J17 shall remain IP67 after immersion in 1 m of water for 30 minutes.

REQ-NVA-ENV-002 | Vibration endurance

The enclosure shall withstand 30 g RMS random vibration for 8 hours per axis without loss of electrical continuity.

REQ-NVA-ENV-003 | Mechanical shock

Following three 50 g half-sine shocks on each axis, no fastener shall loosen and the enclosure shall show no crack longer than 2 mm.

3. Verification-oriented requirements

Identifier	Requirement title	Normative requirement
REQ-NVA-ENV-004	Salt-mist resistance	After 96 hours of salt-mist exposure, connector contact resistance shall remain below 15 mOhm, sealing surfaces shall show no red corrosion, and labels shall remain legible.
REQ-NVA-ENV-005	Mounting integrity	Each of the four mounting bolts shall be tightened to 22 Nm, marked with torque witness paint, and retain at least 18 Nm after vibration.

Applicability notes

Context	Interpretation
Unless a requirement states otherwise	Limits apply to the complete production-intent assembly.
Use of shall	The complete sentence is normative; table columns do not split obligations.

4. Normative installation figure

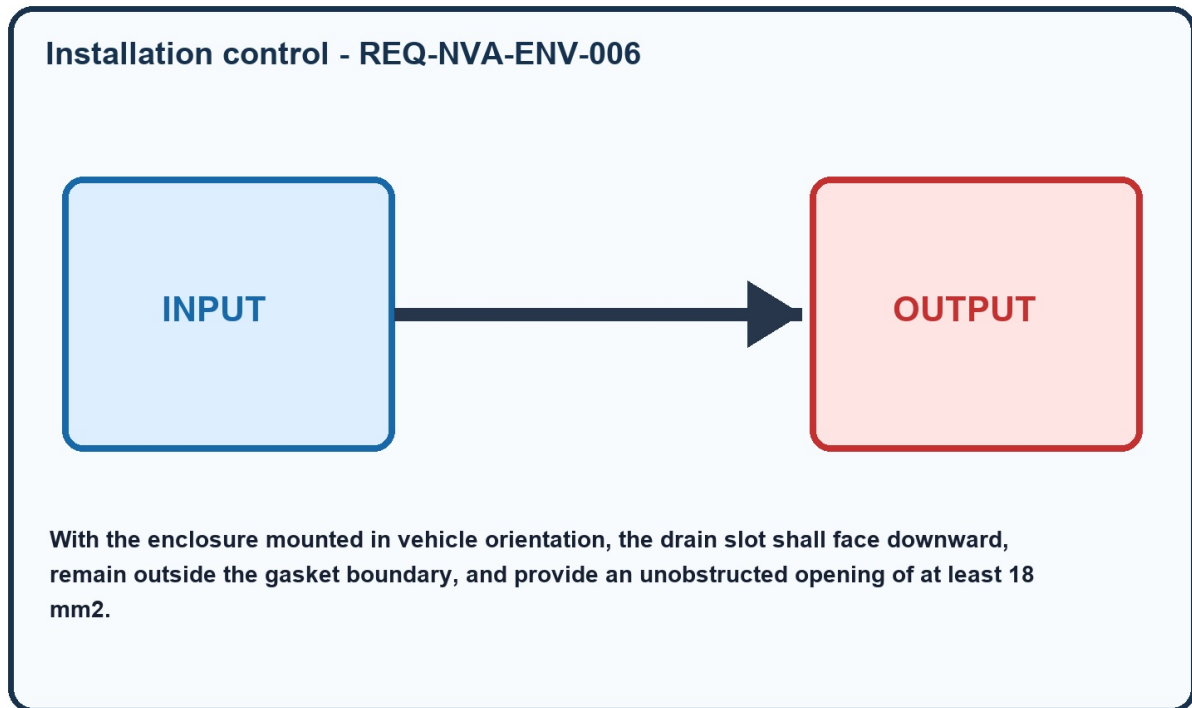


Figure 4-1. Colored geometry and embedded annotation are normative.

Engineering note

The figure annotation forms part of the controlled requirement baseline.

Color, direction, sequence, and geometric relationships must be interpreted visually.